MYDFIR Wazuh SOC Investigation Report

Controlled Security Exercise • Wazuh SIEM / Active Response / Account & File Monitoring

Environment	Windows VM, Ubuntu VM, and Wazuh monitoring server
Activity	Simulated SSH brute-force activity, account changes, and file creation/deletion
Date	September 8, 2026
Severity	Controlled lab activity — no real-world compromise

1. Findings

- Three failed SSH authentication attempts were generated from the Windows VM against the Ubuntu VM.
- An automated Wazuh Active Response was configured to respond after three failed SSH attempts.
- Wazuh recorded Windows account-management activity, including the **Guest** account being enabled and subsequent account-change events.
- A test file was created and deleted as part of the exercise, with file activity monitored by Wazuh.
- The Wazuh dashboard successfully centralized and visualized the authentication, account, and security events generated during the investigation.

2. Investigation Summary

A controlled security exercise was performed in a Wazuh homelab to simulate suspicious authentication and endpoint activity. Three consecutive failed SSH login attempts were generated from the Windows VM toward the Ubuntu VM. An Active Response rule was configured to automatically respond after the third failed attempt, demonstrating detection and automated containment.

Additional endpoint activity was generated to test account and file monitoring. Wazuh recorded the Windows Guest account being enabled, account-change events, and file creation/deletion activity. The events were then reviewed through the Wazuh dashboard and event search.

The activity was intentionally generated for the lab and is not evidence of an external compromise. The exercise demonstrated the ability to detect, investigate, and respond to suspicious activity using centralized SIEM telemetry.

3. Who, What, When, Where, Why, How

Who	The activity involved the test accounts and hosts within the lab. The Windows Guest account was specifically observed being enabled.
What	Three failed SSH authentication attempts were generated, followed by the configured automated response. Windows account-management and file activity was also generated and detected.
When	September 8, 2026, with Wazuh events shown around 17:57–17:58. The report uses UTC+8 (Philippine Time) for the lab timeline. The activity was part of the controlled exercise and was not ongoing after testing.
Where	The activity occurred between the Windows VM and Ubuntu VM monitored by the Wazuh server. Wazuh also recorded authentication activity associated with source IP 192.168.56.101.
Why	The activity was intentionally generated to test Wazuh detection, monitoring, investigation, and automated response capabilities.
How	Failed SSH authentication attempts were manually generated from the Windows VM. Wazuh Active Response was configured to react after three failures. Additional Windows account and file activity was then generated and investigated through Wazuh.

4. Recommendations

- Continue monitoring repeated authentication failures and investigate patterns that may indicate brute-force activity.
- Maintain an Active Response control for repeated SSH failures, while tuning thresholds to reduce false positives.
- Monitor account enable/disable, creation, deletion, and privilege changes, especially for unexpected or unauthorized activity.
- Continue using File Integrity Monitoring to detect unexpected file creation, modification, and deletion on monitored endpoints.
- Review Wazuh alerts and custom rules regularly to ensure detections are accurate and actionable.
- Document and validate automated response actions in a controlled environment before deploying equivalent controls to production systems.

Conclusion: The exercise demonstrated a complete basic SOC workflow: generate activity, detect it with centralized telemetry, investigate the events, and apply an automated response. Wazuh successfully provided visibility into authentication, account, and file activity within the lab environment.